

BASIC LINUX FOR LIFE SCIENCES

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Learning Outcomes

At the end of this, you should be able to:

1. host Linux in windows.
2. understand the command line interface;
3. understand structure of files and directory;
4. create, move and access files;
5. read files and extract needed information;
6. concatenate commands and direct outputs

Hosting Linux on Windows



VirtualBox

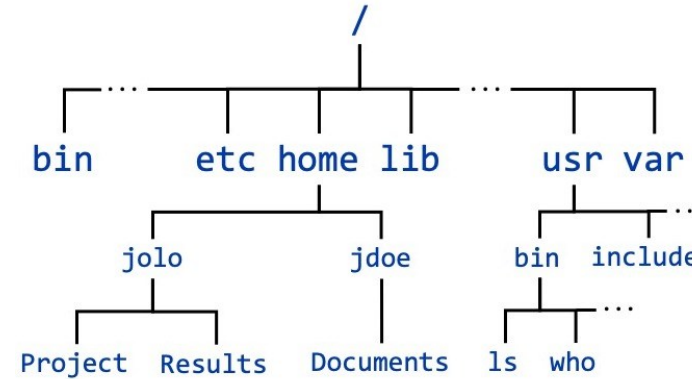
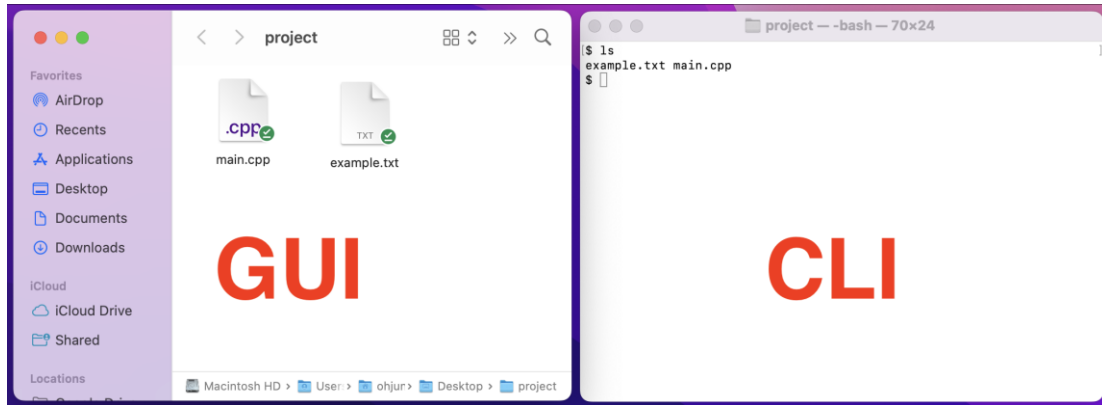


```
Command Prompt x Windows PowerShell x + v
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\shadr> wsl --install|
```

Command Line Interface and File Structure



- **Directory:** hierarchical organization of folders and files
- **Root directory (/):** All other directory are built on this
- **Home directory (~):** Landing page when you open the terminal
- **Working directory (.):** your current directory

- **Path:** locates a given file in the system hierarchy
- **Absolute path:** path from root
- **Relative path:** path from current directory

Basic Commands

<code>pwd</code>	Prints the absolute path of current directory
<code>cd</code>	Change from one directory to another
<code>ls (-a, -l, -al)</code>	List content of a directory (plus the flags to use)
<code>mkdir</code>	Creates new directory
<code>rm (-r)</code>	Remove a file (or a directory with the <code>-r</code> flag)
<code>cp</code>	Copy command
<code>mv</code>	Move command (also used to rename files/folders)
<code>vi (vim, nano, gedit)</code>	To visualize file (text files)
<code>cat (less, more, head, tail)</code>	To print out the contents of a file
<code>wc</code>	Word count; outputs character, line and word counts
<code>grep</code>	To search for a pattern in a file
<code>chmod</code>	Change permissions of files
<code>wget</code>	To get something from the web using its link

Copy Move Rename and Remove

- Copy a file from one destination to another

cp <path_from/file> <path_to>

- To move a file from one location to another.
The file is erased from the original location

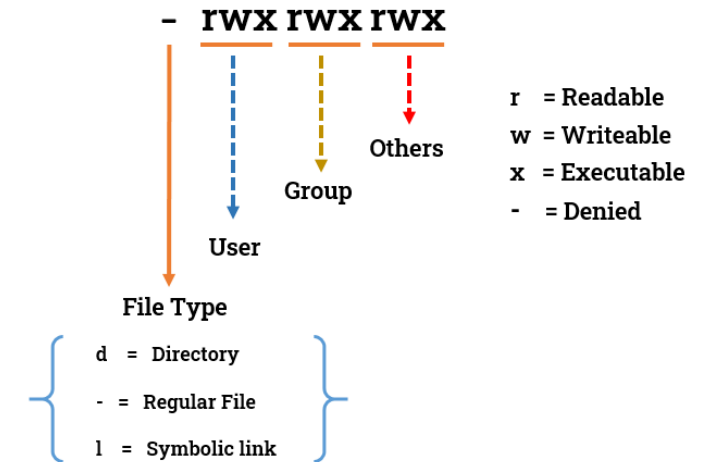
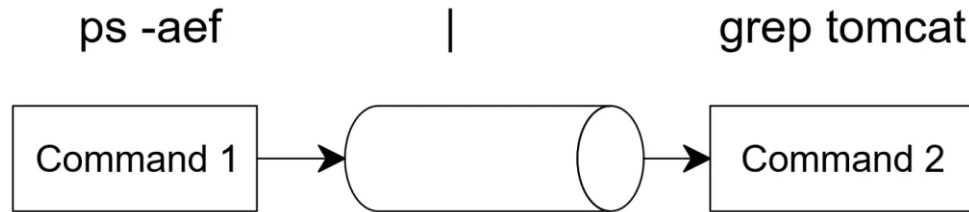
mv < path_from /file> <path_to>

- To rename a file in same directory

mv <path_from/file> <sme_path/new_file_name>

- Use “>” command to send the contents of one file or output of a command into another file
- Use “> >” to do same but the new file will be created or existing one will be overridden
- cat smiles.pdb > maga.txt
- cat smiles.pdb > > maga.txt

Combining commands and file permissions



- Use “|” command to send the output of one command as input of the next
- Use “;” to combine commands sequentially
- Use “&&” to combine commands. The next commands works only if the previous one succeeds

- Use “ls -l” command to see permissions
- File permissions are important
- Use “chmod -x (or -w or -r)” to change permission
- Only users can change file permissions

Use of Wildcards

Wildcards are a group of **special characters** which allow filenames to be selected based on pattern of characters

Example: Specific file extension

This example searches for all files with the extension ".txt" in the current directory and passes them to the ``ls`` command. The found filenames are then displayed in the terminal.

Input:

```
ls *.txt
```

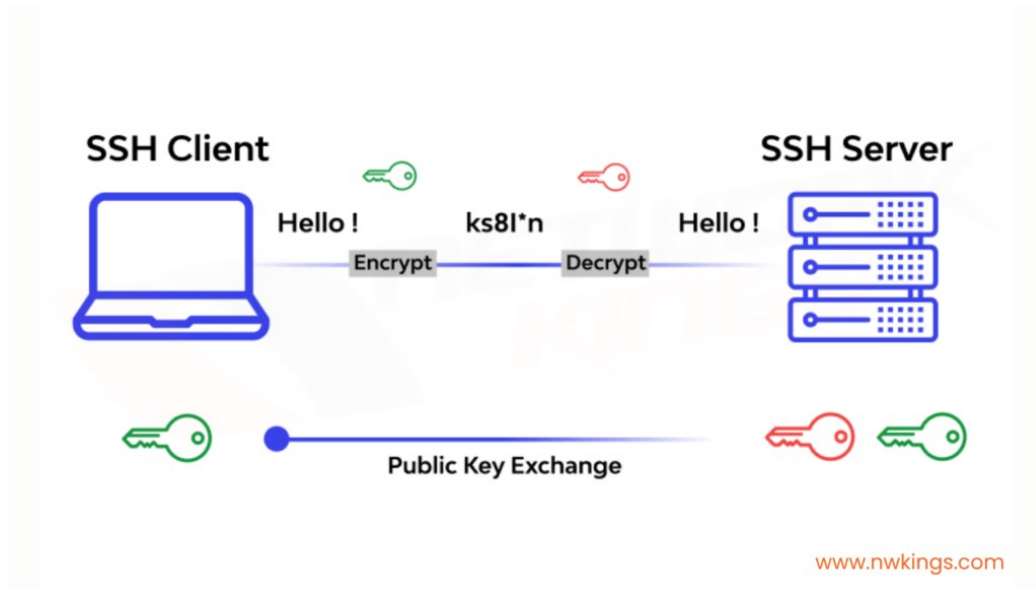
Output:

```
command-find.txt
command-grep.txt
linux-authorization-system.txt
```

Wildcard	Meaning
*	Matches any characters in a filename
?	Matches any single/one character
[!characters]	Matches any character that is not a member of the set characters
[characters]	Matches any character that is a member of the set <i>characters</i> . The set of characters may also be expressed as a <i>POSIX character class</i> such as one of the following: [:alnum:] Alphanumeric characters [:alpha:] Alphabetic characters [:digit:] Numerals [:upper:] Uppercase alphabetic characters [:lower:] Lowercase alphabetic characters

Wildcard	Meaning
a*	Any file name starting with a
*	All possible filenames
A*.fasta	All filenames that begin with A and end with .fasta
????vcf	Any filenames that contain exactly 4 characters and end with .vcf
[abc]*	Any filename that begins with "a" or "b" or "c" followed by any other characters
[:upper:]*	Any filename that begins with an uppercase letter. This is an example of a character class

SSH (Secure Shell)



- Enables secure system administration and file transfers over insecure networks through a protocol
- SSH is very important to connect to high computing clusters for many simulation/drug design jobs
- There are many ssh client software based on GUI
- SSH keys can help automate access to servers but must be managed carefully

SSH (On the Terminal)

```
$ ssh jolo@foo.edu
```

Connect to a remote server; prompts for password

```
$ ssh -i my_key jolo@foo.edu
```

Connect to a remote server; uses a private key

```
$ scp code.sh jolo@foo.edu:~/Project
```

Copy a file (code.sh) from current directory to a remote server

```
$ scp jolo@foo.edu:~/output.txt .
```

Copy a file (output.txt) from a remote server to current directory



NOTES ABOUT LINUX

- No real distinction between the names of ordinary files and the names of directory files.
- No two files in the same directory can have the same name.
- Files in different directories can have the same name.
- Linux is case-sensitive: `course_data`, `course_Data` and `Course_data` are different and would represent three distinct files or folders.

EXTRA RESOURCES

- <https://ubuntu.com/tutorials/command-line-for-beginners#1-overview>
- <https://www.hackthebox.com/blog/learn-linux>
- <https://www.freecodecamp.org/news/linux-command-line-tutorial/>
- <https://www.youtube.com/watch?v=16d2lHc0Pe8>
- <https://www.youtube.com/watch?v=16d2lHc0Pe8&vl=en>
- <https://cvw.cac.cornell.edu/linux/intro/index>